MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, RECEPTACLE, DECADE INCREMENT 20 THROUGH 110 AND 128 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.050 SPACING)

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors decribed herein shall consist of this specification and the latest issue of MIL-C-55302.

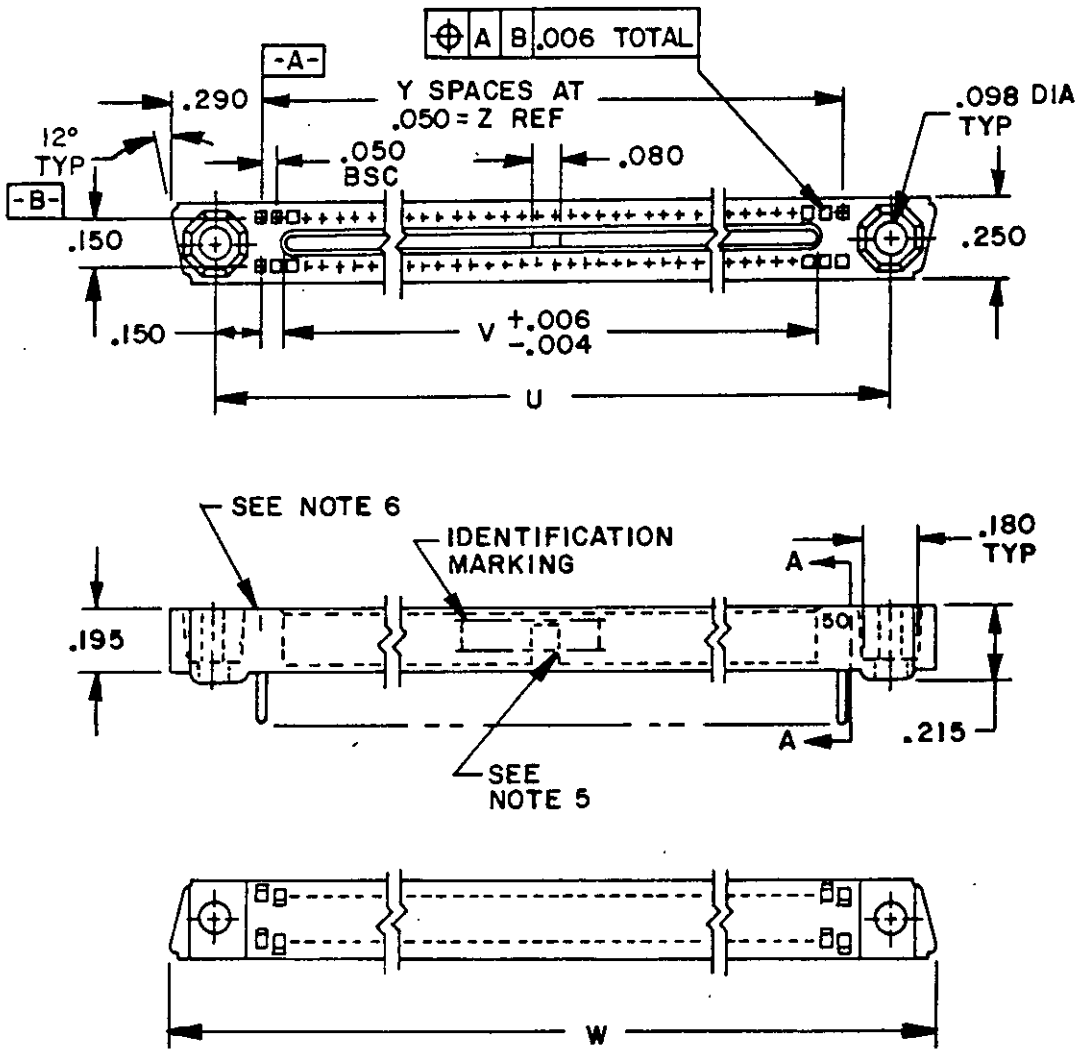


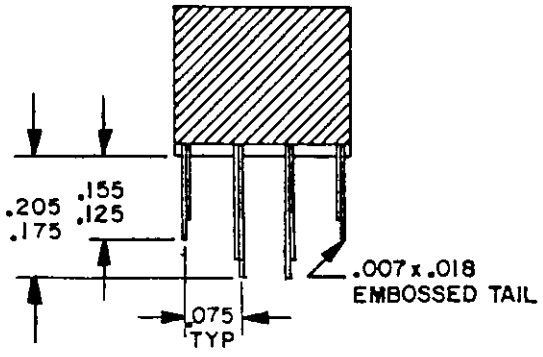
FIGURE 1. Connectors, receptacle .050 (1.27 mm) spacing.

(E) denotes change.

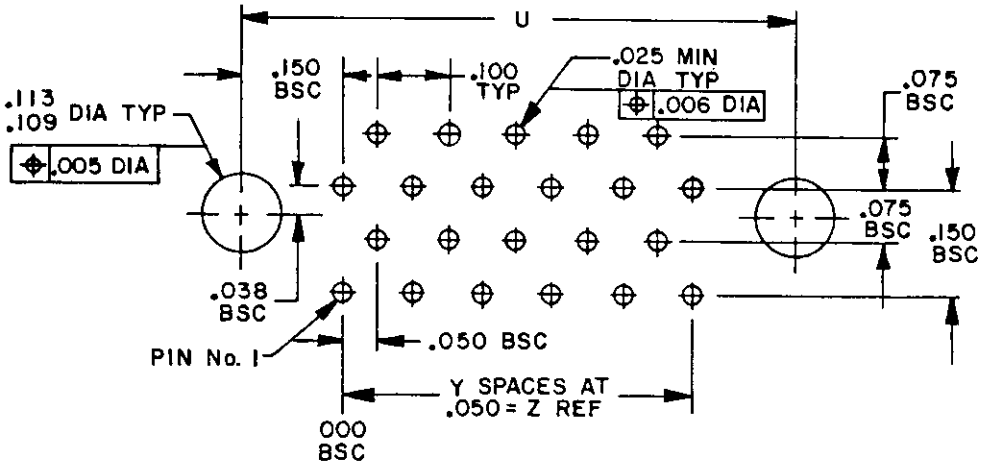
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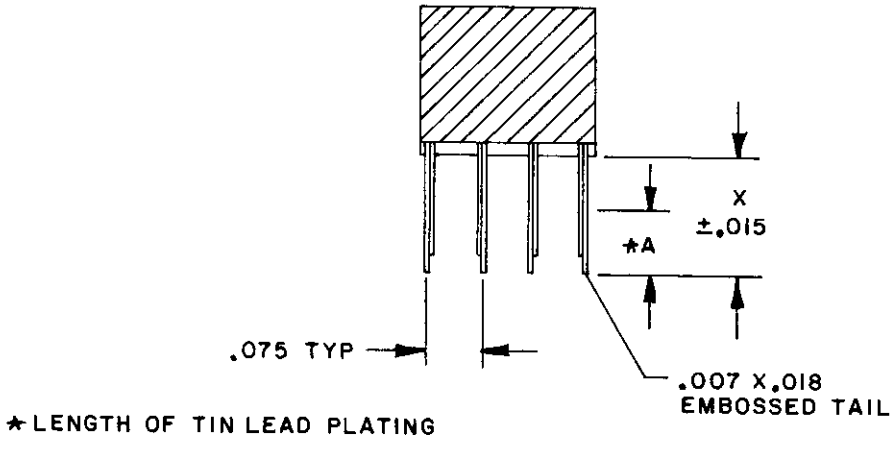
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SECTION A–A
TYPE I



RECOMMENDED PRINTED WIRING BOARD HOLE
PATTERN, CONNECTOR SIDE



★LENGTH OF TIN LEAD PLATING

SECTION A–A
TYPE II

FIGURE 1. Connectors, receptacle .050 (1.27 mm) spacing - Continued.

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(E) TABLE I. Dash numbers and dimensions.

Type	Dash number	No. of contacts	Dimensions							Contact identification numbers
			A	U	V	W	X	Y	Z	
I	01	20		.750 (19.05)	.300 (7.62)	1.030 (26.16)		9	.450 (11.43)	1-10
I	02	30		1.000 (25.40	.550 (13.97)	1.030 (32.51)		14	.700 (17.78)	1-15
I	03	40		1.250 (31.75	.800 (20.32)	1.530 (38.86)		19	.950 (24.13)	1-20
I	04	50		1.500 (38.10	1.050 (26.67)	1.780 (45.21)		24	1.200 (30.48)	1-25
I	05	60		1.750 (44.45	1.300 (33.02)	2.030 (51.56)		29	1.450 (36.83)	1-30
I	06	70		2.000 (50.80	1.550 (39.37)	2.280 (57.91)		34	1.700 (43.18)	1-35
I	07	80		2.250 (57.15	1.800 (45.72)	2.530 (64.26)		39	1.950 (49.53)	1-40
I	08	90		2.500 (63.50	2.050 (52.07)	2.780 (70.61)		44	2.200 (55.88)	1-45
I	09	100		2.750 (69.85	2.300 (58.42)	3.030 (76.96)		49	2.450 (62.23)	1-50
I	10	110		3.000 (76.20	2.550 (64.77)	3.280 (83.31)		54	2.700 (68.58)	1-55
I	11	128		3.450 (87.63	3.000 (76.20)	3.730 (94.74)		63	3.150 (80.01)	1-64
II	12	100	.210 (5.33)	2.750 (69.85	2.300 (58.42)	3.030 (76.96)	.300 (7.62)	49	2.450 (62.23)	1-50
II	13	110	.210 (5.33)	3.000 (76.20	2.550 (64.77)	3.280 (83.31)	.300 (7.62)	54	2.700 (68.58)	1-55
II	14	128	.130 (3.30)	3.450 (87.63	3.000 (76.20)	3.730 (94.74)	.190 (4.82)	63	3.150 (80.01)	1-64

FIGURE 1. Connector, receptacle, .050 (1.27 mm) spacing - Continued.

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Inches	mm	Inches	mm	Inches	mm
.004	0.10	.050	1.27	.150	3.81
.005	0.13	.075	1.90	.155	3.94
.006	0.15	.080	2.03	.175	4.44
.007	0.18	.098	2.49	.180	4.57
.015	0.38	.100	2.54	.195	4.95
.018	0.46	.109	2.77	.205	5.21
.025	0.64	.113	2.87	.215	5.46
.038	0.97	.125	3.18	.250	6.35
				.290	7.37

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) on decimals and 2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/117(USAF) and MIL-C-55302/118(USAF).
5. Except 20 through 50 position.
6. Numbers indicating end positions stamped this side only.

FIGURE 1. Connectors, receptacle .050 (1.27 mm) spacing - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figure 1.

Plating (contact): Contact plating shall be gold in accordance with MIL-G-45204, type II, grade C, class I over nickel in accordance with QQ-N-290, class 2, 30 to 150 microinches overall. If the selective plating process is used, contact plating shall be gold in accordance with MIL-G-45204, type II, grade C, class I over nickel in accordance with QQ-N-290, class 2, on the contact engagement area for .100 inch minimum length. The termination area plating shall be tin lead in accordance with MIL-P-81728, class I over nickel in accordance with QQ-N-290, class 2, 30 to 150 microinches for .300 inch minimum length.

Contact identification: See figure 1.

Oversized pin exclusion: .0330 diameter pin.

Keying: See MIL-C-55302/31.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by ,30 and the withdrawal force in pounds shall be a minimum of .30 times the number of contacts and shall not exceed the measured insertion force.

Contact engagement and separation force:

Engagement force: Maximum force to engage maximum test pin .0190 +.0000, -.0001 diameter shall be 6 ounces.

Separation force: Minimum force to extract minimum test pin .0170 +.0000, -.0000, diameter shall be .10 ounces.

Contact rating: 1.5 amperes per contact, maximum.

Contact resistance: When mated with MIL-C-55302/117, the average contact resistance of all contacts measured shall not exceed .025 ohm, and no individual contact pair shall have a resistance exceeding .030 ohm. When mated with MIL-C-55302/118, the average contact resistance of all contacts measured shall not exceed .015 ohm, and no individual contact pair shall have a resistance exceeding .020 ohm.

Contact retention: .75 pound, minimum.

Operating temperature: -65°C to +125°C.

Dielectric withstanding voltage:

Sea level: 600 volts rms.

High altitude: 150 volts rms.

Solderability: Each connector shall be subjected to method 208 of MIL-STD-202 (solder termination only) (applicable to qualification and retention of qualification only)

Part number: M55302/119- (dash number from table I).

Patent number 3,404,367. The Government has a royalty free licence under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

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Air Force - 19

Preparing activity:
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Agent:
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(Project 5935-F282)

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